

UWS 2 Documentation

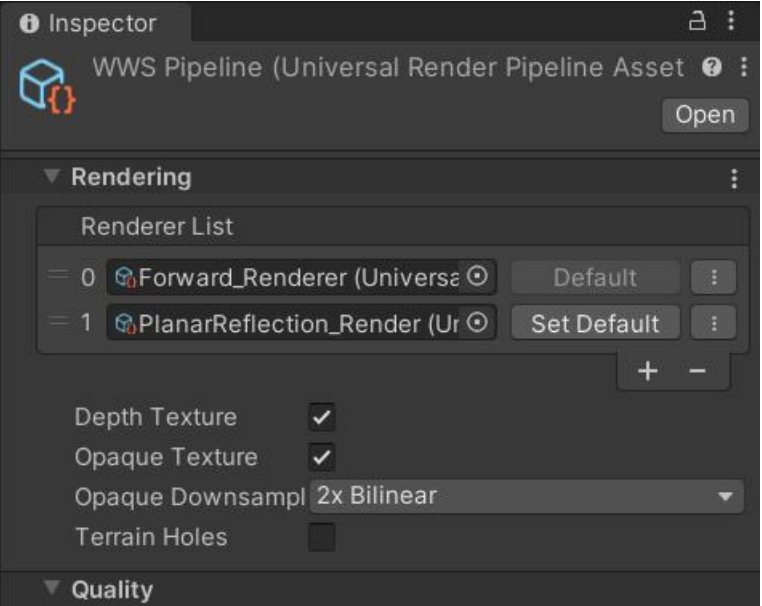


Water

1.1 Setup

Before starting, ensure that your project is in the URP environment.

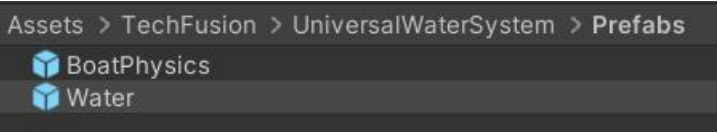
In your current Universal Render Pipeline Asset, add the renderer at index 1 to the Renderer List. This is for preparing Planar Reflection. Then, check Depth Texture and Opaque Texture, as shown in the following picture.



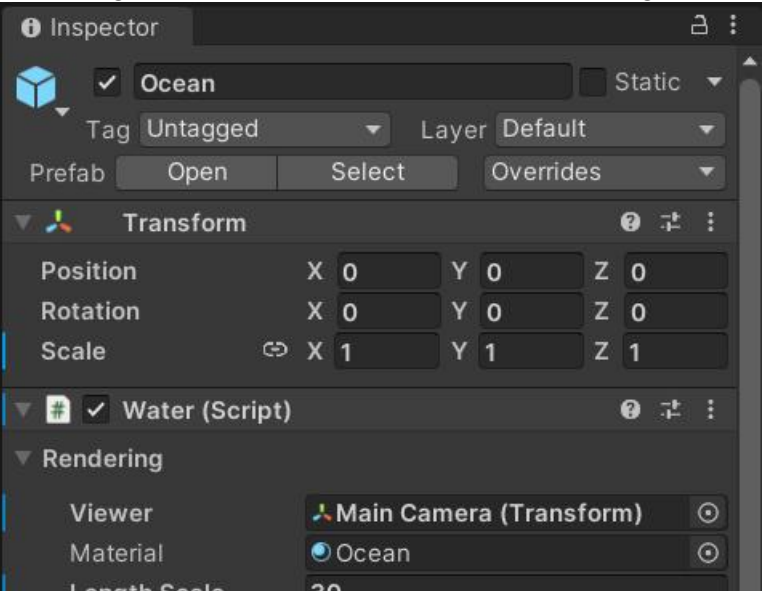
Then, add a Water Renderer Feature to renderer 0.



Setting up UWS is simple. Drag the Water.prefab file from the Prefabs folder into the scene.

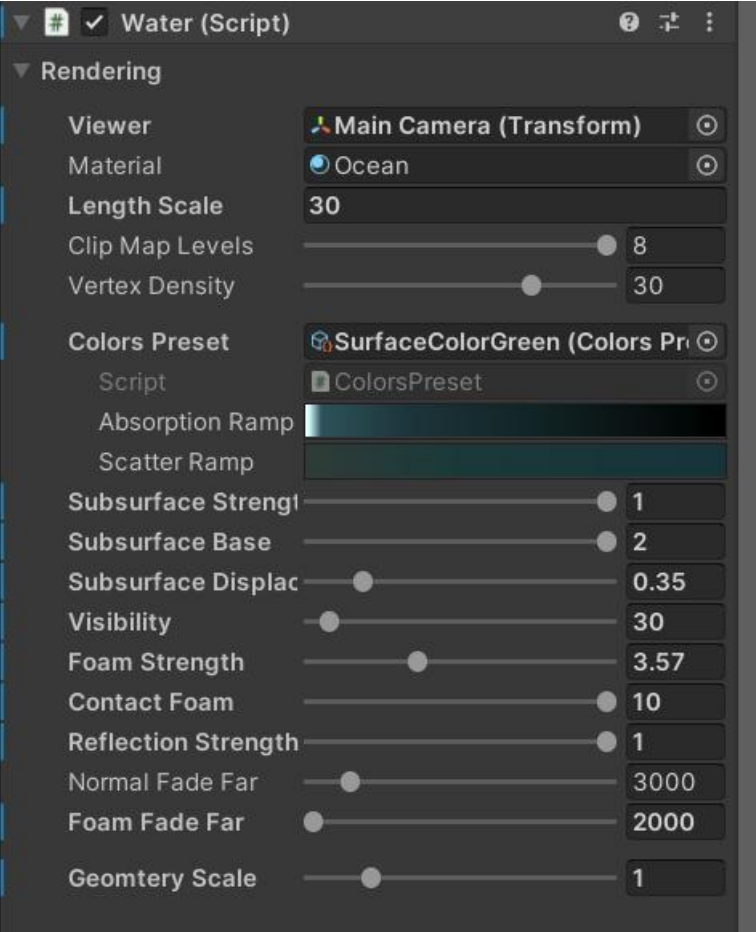


Then, drag the camera into the Viewer under Rendering in the Inspector, and you're ready to go.



1.2 Parameters

Rendering Parameters



- **Viewer:** Camera Transform for rendering.
- **Material:** Material.
- **Length Scale:** Controls the size of the generated LOD Mesh.
- **Clip Map Levels:** Controls the number of LOD Mesh layers.
 - **Vertex Density:** Controls vertex density.
 - **Colors Preset:** Preset color values.
- **Absorption Ramp:** Controls color absorption at different depths.
 - **Scatter Ramp:** Controls light scattering at different depths.
 - **Subsurface Strength:** Subsurface scattering intensity.
- **Subsurface Base:** Subsurface scattering base value.
- **Subsurface Displacement Scale:** Increase in subsurface scattering based on wave height.
- **Visibility:** Water visibility.
- **Foam Strength:** Controls foam amount.
- **Contact Foam:** Controls contact foam strength.
- **Reflection Strength:** Reflection strength.
- **Normal Fade Far:** Distance at which normals become vertical to the ground.
- **Foam Fade Far:** Distance at which foam dissipates in the distance.

Simulation Parameters

Simulation

FFT Level

7

Update Every Fram

☐

Wave Settings

MobileModerate (Waves Setti

Script

WavesSettings

G

15

Depth

100

Wind Speed

9

Local

Scale

1

Wind Direction

0

Tile Strength

1000

Spread

0.611

Swell

1

Peak

2

Wave Length F

0.01

Swell

Scale

1

Wind Direction

105

Tile Strength

173107

Spread

0.436

Swell

0.311

Peak

1.34

Wave Length F

0

Far Fade

250

Middle Fade

50

Near Fade

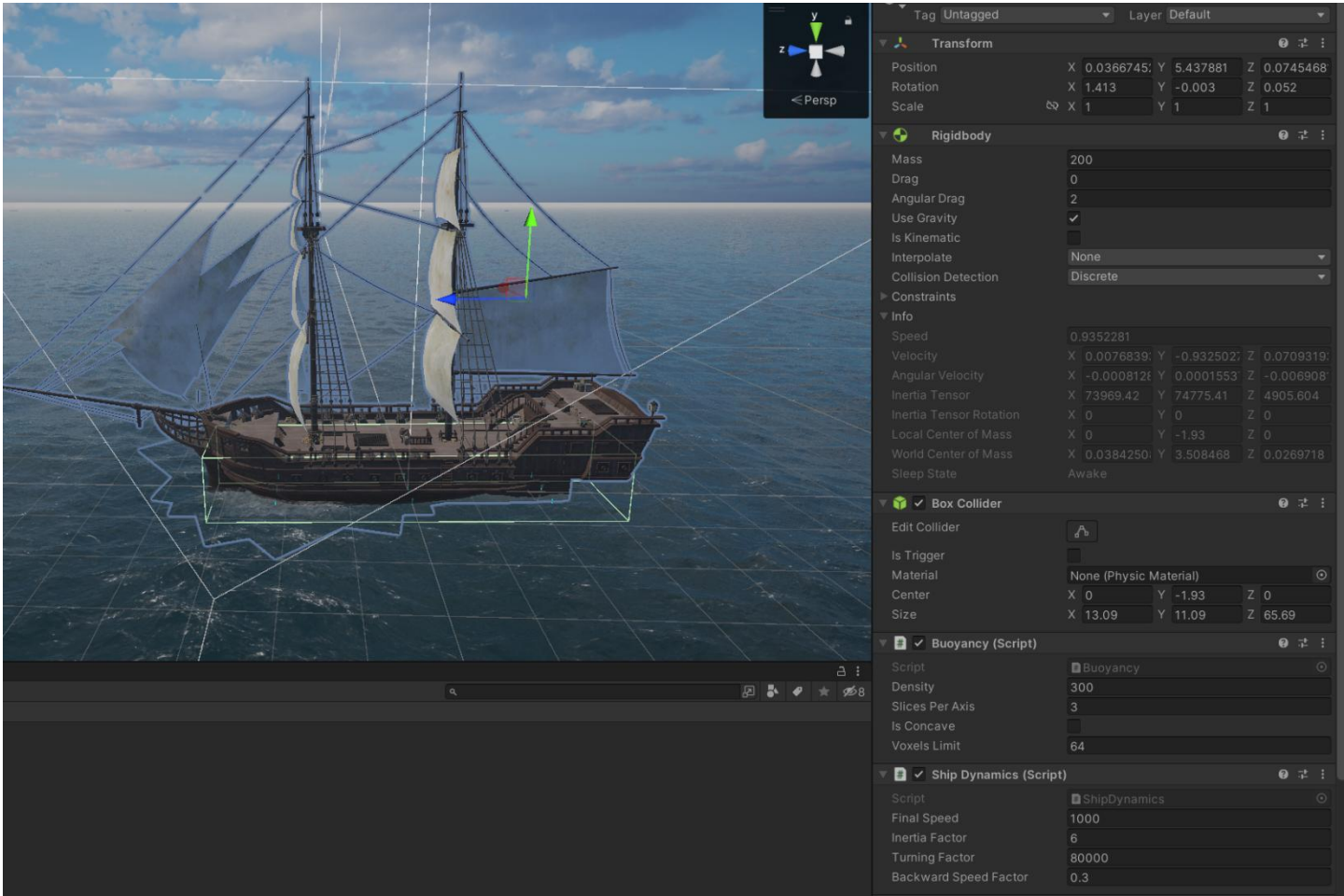
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- **FFT Level:** FFT calculation resolution level.
- **Update Every Frame:** Dynamically change wave shape properties at runtime.
- **Wave Settings:**
 - **G:** Gravity acceleration.
 - **Depth:** Simulated water depth.
 - **Wind Speed:**
 - Local:** Detail waves.
 - Swell:** Overall waves.
 - **Scale:** Applied scale for waves.
 - **Wind Direction:** Controls the plane angle of the wind.
 - **Tile Strength:** Controls Tiling intensity (smaller values result in tighter tiling).
 - **Spread:** Controls the influence of wind direction on water waves.
 - **Swell:** Controls wave detail.
 - **Peak:** Controls wave peak intensity.
 - **Wave Length Filter:** Filters wavelengths smaller than this value.
- **Far Fade:** Outer layer decay distance in the three-layer FFT.
- **Middle Fade:** Middle layer decay distance in the three-layer FFT.
- **Near Fade:** Inner layer decay distance in the three-layer FFT.

Boat

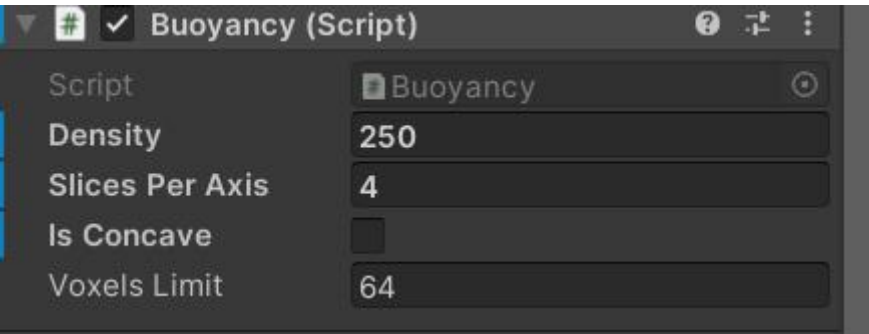
2.1 Buoyancy

Refer to the BoatPhysics in the Sailing Scene.



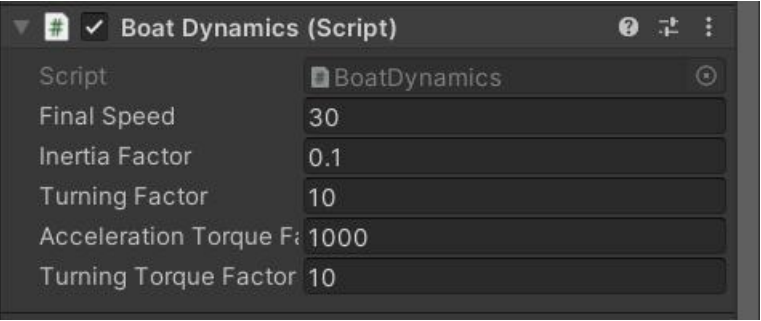
You Need Rigidbody and BoxCollider components on the GameObject

1. Buoyancy component with the following parameters:



- **Density:** Density of the ship's hull; smaller values result in greater buoyancy.
- **SlicesPerAxis:** Density of buoyancy calculation, recommended 2~4.
- **IsConcave:** Set to `True` for more accurate calculations when the Collider with the "Ship" tag is a MeshCollider (default is `False`).
- **VoxelsLimit:** Maximum number of calculation points for buoyancy.

2.2 BoatDynamics

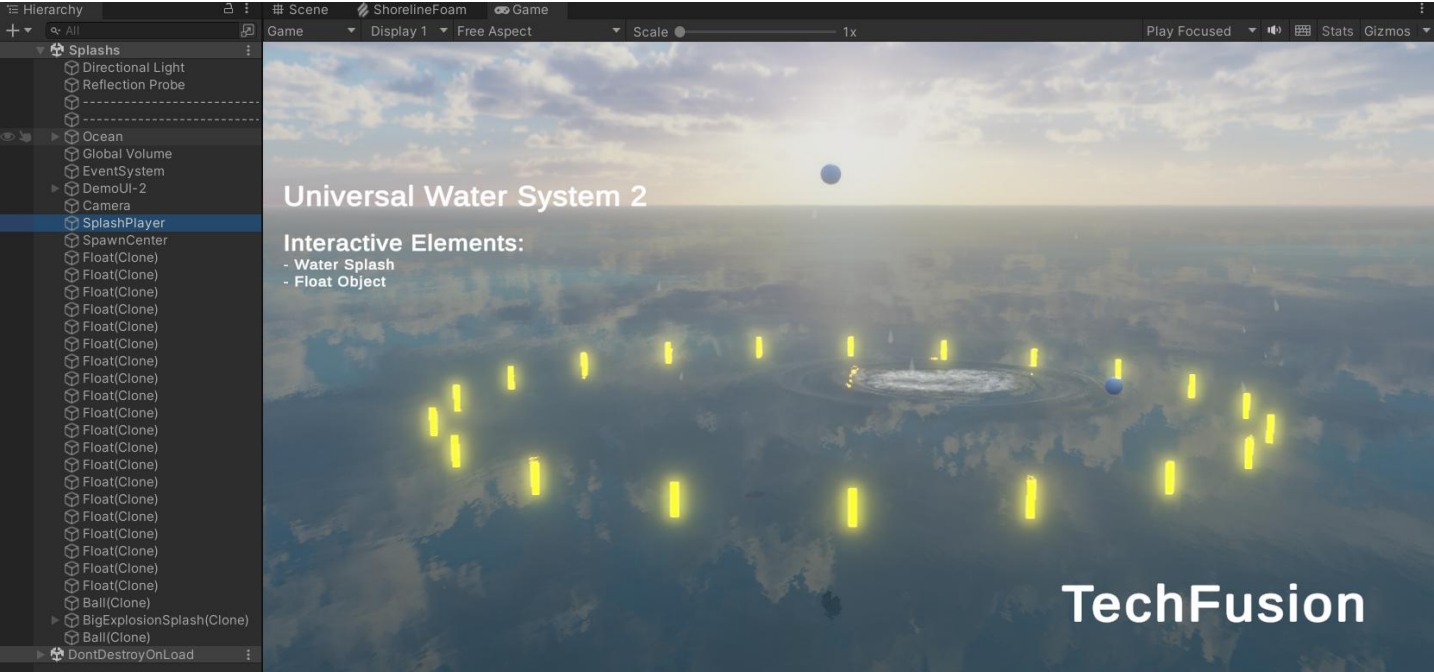


- **Final Speed:** Maximum traveling speed.
- **Inertia Factor:** Coefficient for speed change.
- **Turning Factor:** Torque coefficient for turning.
- **backwardSpeedFactor:** Ratio of backward speed to forward speed.

Interaction

3.1 Collision

Refer to the SplashPlayer in the Splashes Scene.



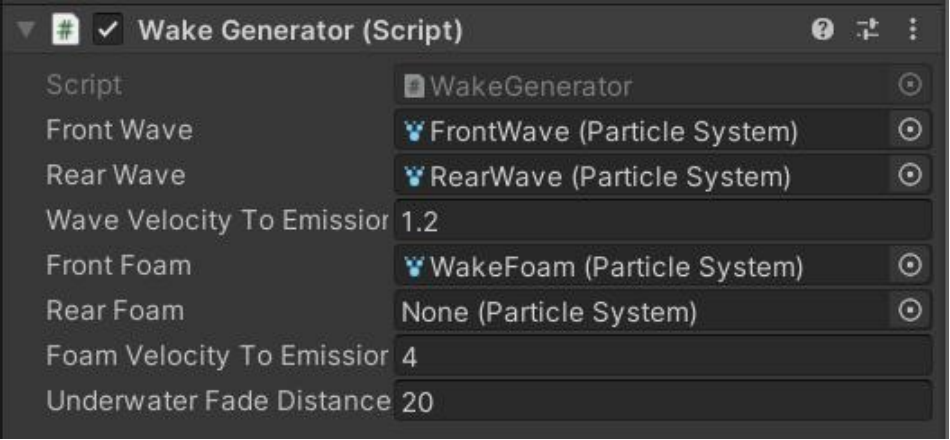
The working principle of SplashPlayer is to first instantiate a Ball and then check if the Ball falls below the water surface.

```
float waterHeight = Water.Instance.GetWaterHeight(transform.position);  
  
if (waterHeight > transform.position.y)  
{  
  
    //.....  
  
}
```

If it does, a BigExplosionSplash prefab (located at Assets/TechFusion/UniversalWaterSystem/Prefabs/Effect/BigExplosionSplash.prefab) is instantiated at the current location.

3.2 Wake Foam

Refer to the BoatPhysics in the Sailing Scene.

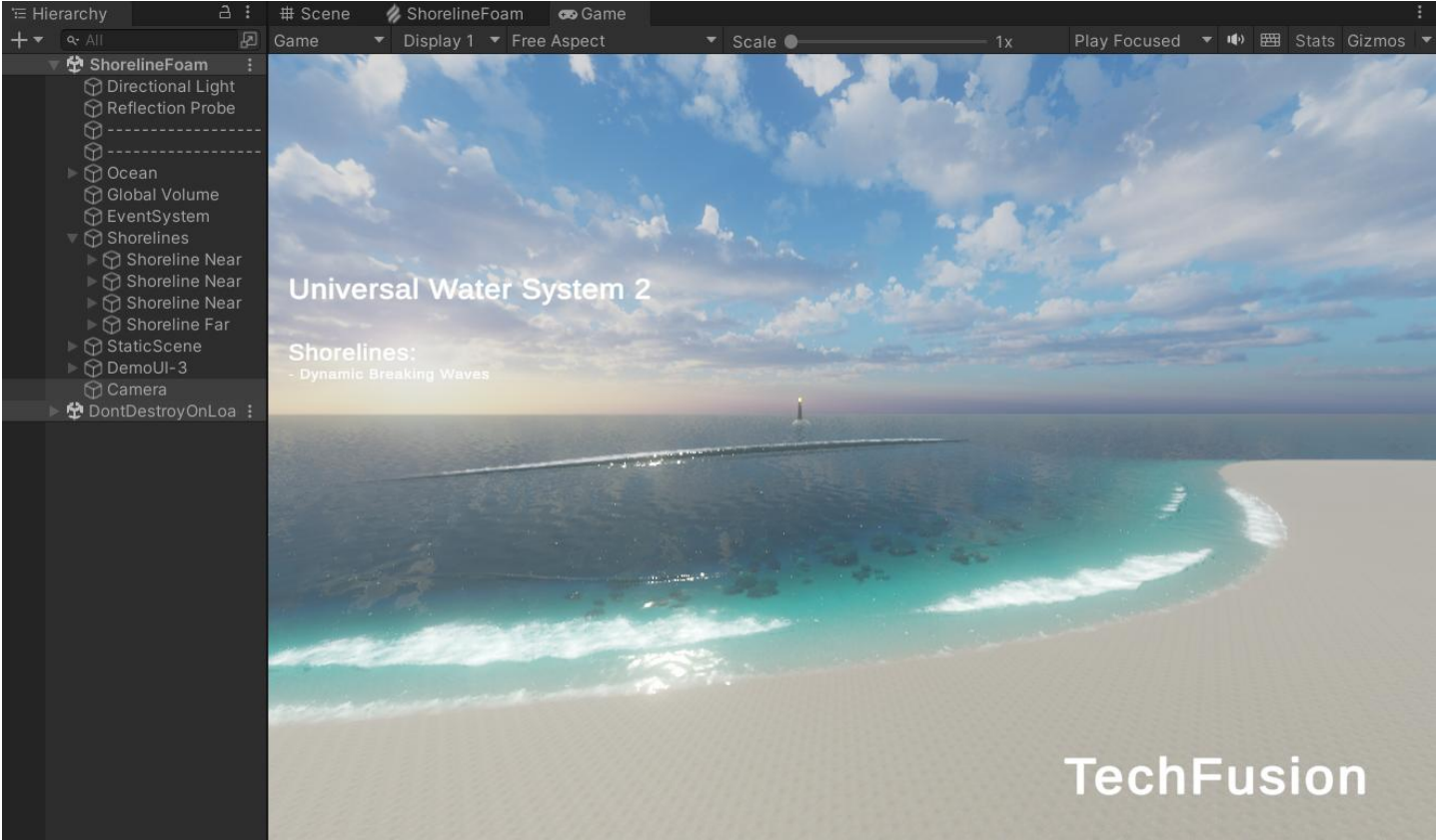


- **FrontWave and RearWave:** Particle emitters for waves at the ship's front and rear.
- **WaveVelocityToEmissionRate:** Impact of ship's movement speed on wave particle emitter emission rate.
- **FrontFoam and RearFoam:** Particle emitters for foam at the ship's front and rear.
- **FoamVelocityToEmissionRate:** Impact of ship's movement speed on foam particle emitter emission rate.
- **UnderwaterFadeDistance:** Maximum fade distance underwater.

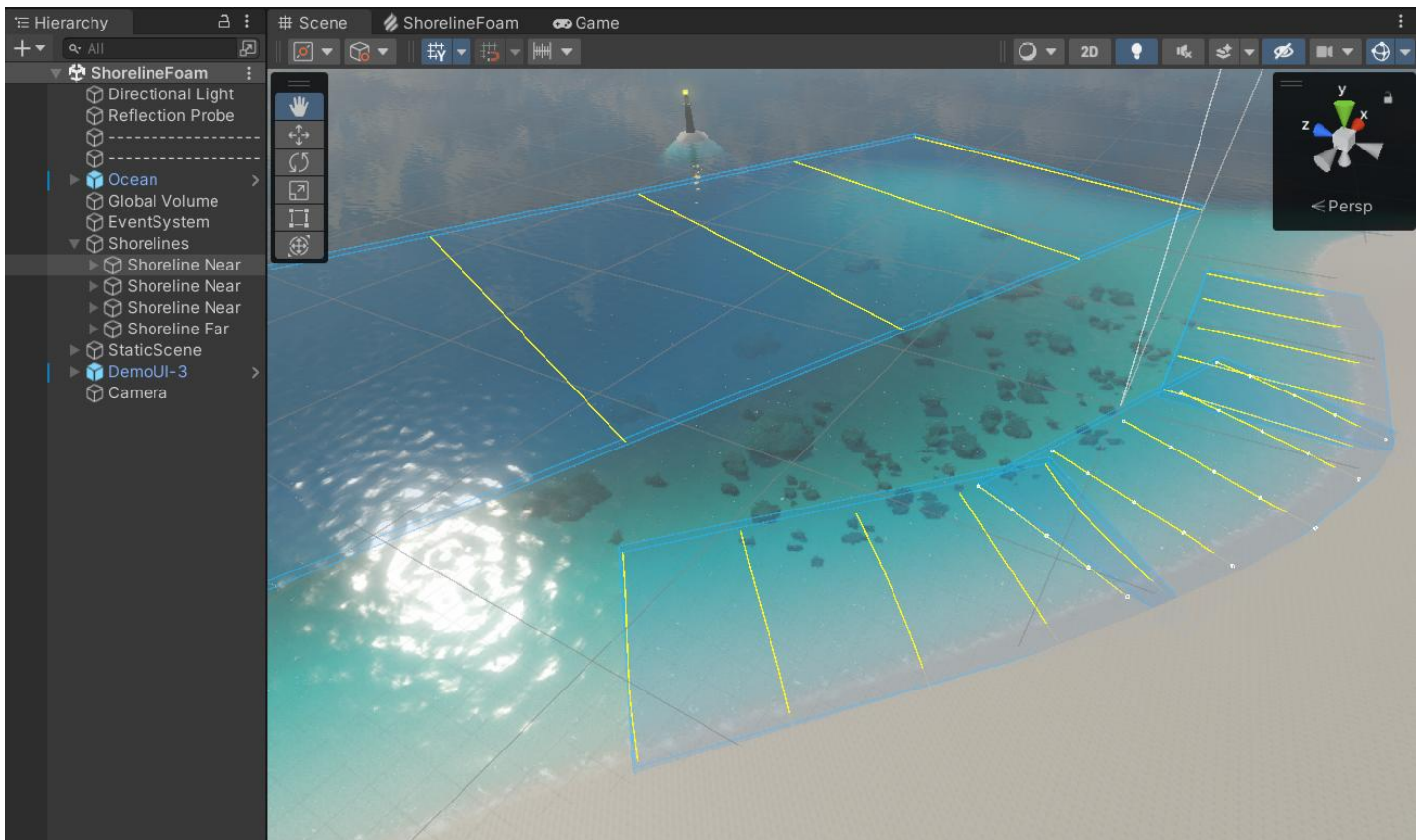
Shoreline

4.1 Guideline

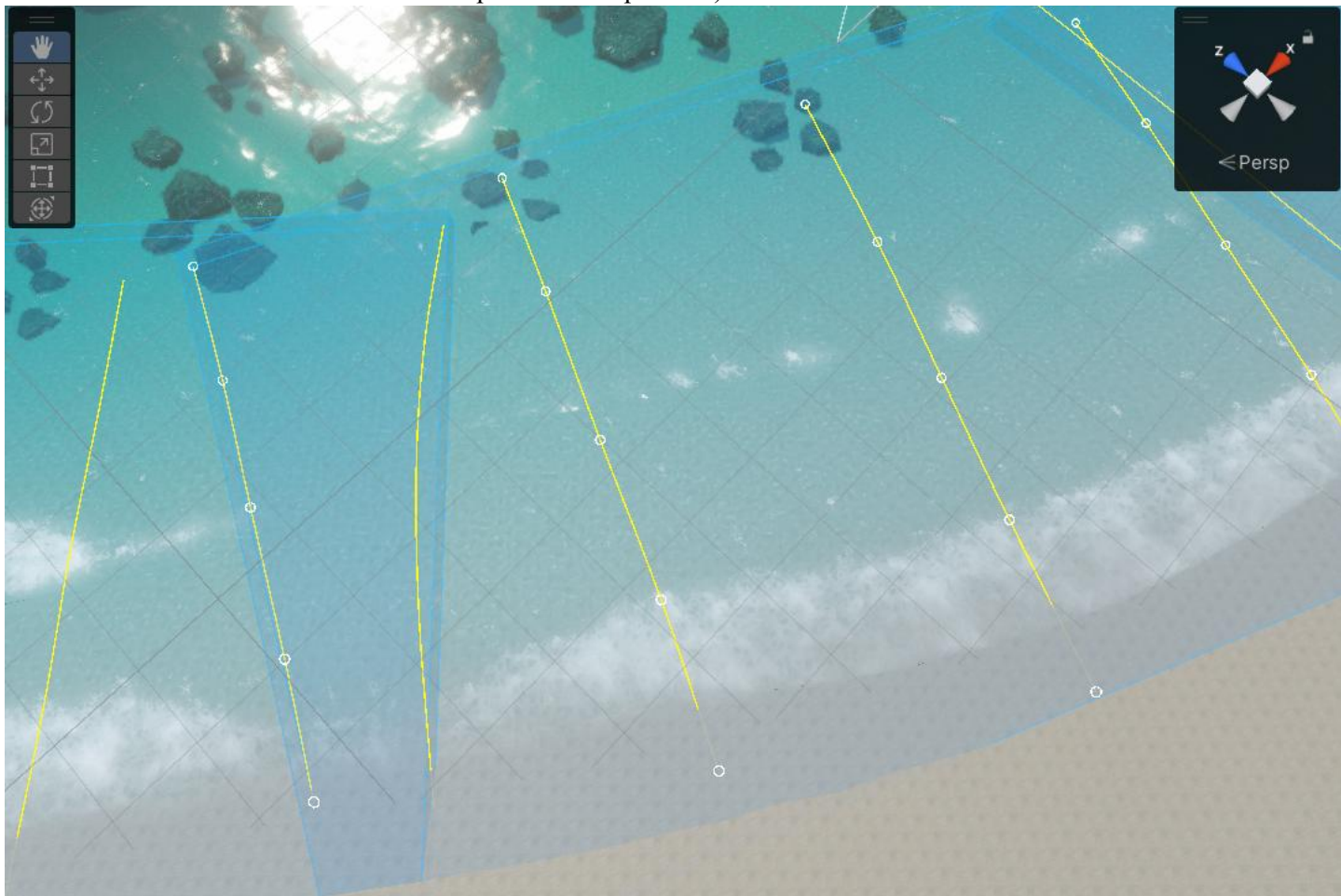
Refer to the Shoreline in the ShorelineFoam Scene.



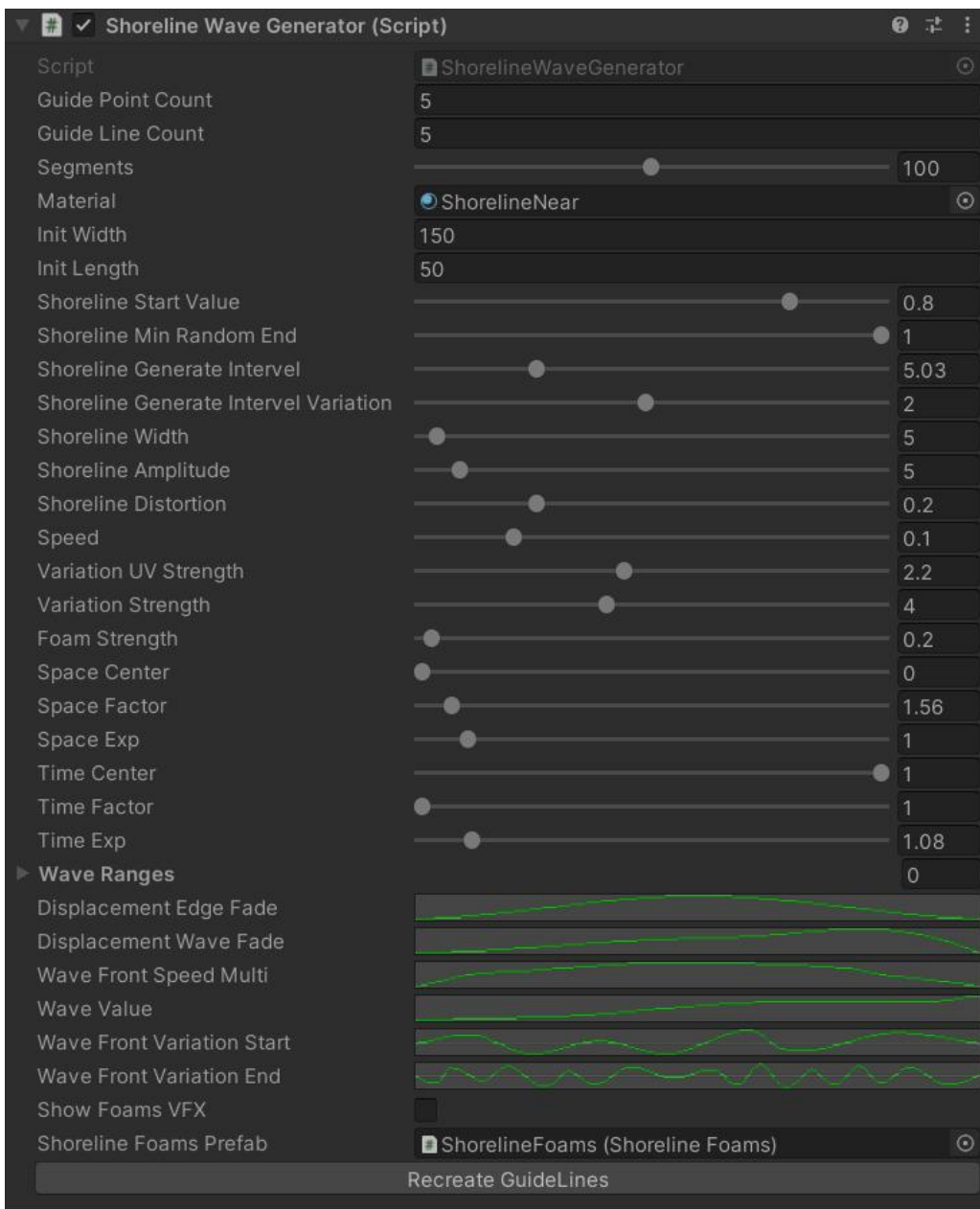
In the Scene, the objects with a light blue border are Shoreline Wave Generators. They control the morphing of the shoreline using Guidelines. These yellow lines represent the Guidelines, and you can adjust parameters to set the number of Guidelines in each shoreline.



When you select a Shoreline Wave Generator object, you will see some white circles on the Guideline, which are the control points of the Guideline. By dragging these control points, you can shape each Guideline (note that it is best to maintain an even distance between control points where possible).



4.2 Parameters (Experimental)



- **Guide Point Count:** The number of control points on each Guideline (you need to click the Recreate button below after modification).
- **Guide Line Count:** Set the number of Guidelines in each Shoreline Wave Generator (you need to click the Recreate button below after modification).
- **Segments:** Control the degree of mesh subdivision for dynamically generated shorelines.
- **Material:** The material of the shoreline.
- **Init Width and Init Length:** Control the size of the horizontal area affected by the shoreline (you need to click the Recreate button below after modification).
- **Shoreline Start Value:** Control the position on the Guideline where the newly generated shoreline mesh is placed.
- **Shoreline Generate Interval:** The time interval between the generation of two shoreline meshes.
- **Shoreline Generate Interval Variation:** A slight random variation in the time interval between the generations of two shoreline meshes.
- **Shoreline Width:** The length of the shoreline mesh along its direction of movement.
- **Shoreline Amplitude:** The maximum height of the shoreline wave.
- **Shoreline Distortion:** Add a degree of twist to the line shape of the shoreline.
- **Speed:** The movement speed of the shoreline wave.
- **Variation UV Strength:** The diversity level of the shoreline wave crests.
- **Variation Strength:** The intensity of the shoreline wave crest diversity.
- **Foam Strength:** The intensity of foam generated by Shoreline Waves.
- **Space Center:** The position of foam generated by shoreline waves relative to the waves themselves.
- **Space Factor:** A range factor used to calculate the generation position.
- **Space Exp:** A sharpening factor used to calculate the generation position.

- **Time Center:**The time position of foam generated by shoreline waves relative to the waves themselves.
- **Time Factor:**A time range factor used to calculate the generation position.
- **Time Exp:**A sharpening factor used to calculate the generation position.

Please note that some parameters of the shoreline are still in the experimental stage and may be documented after possible modifications in subsequent **versions**.